



Girraween High School

2022

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 2

Total Marks: 100

Section 1 (Pages 2 – 5) **10 Marks**

- Attempt Q1 - Q10
- Allow about 15 minutes for this section

General Instructions

- Reading time: 10 minutes
- Working time: 3 Hours
- Write using a black or blue pen
- Board approved calculators may be used
- Laminated reference sheets are provided
- Answer multiple-choice questions by completely colouring in the appropriate circle on your multiple-choice answer sheet.
- Answer questions 11-16 in the appropriate answer booklet and show all relevant mathematical reasoning and/or calculations.

Section 2 (Pages 6-11) **90 marks**

- Attempt Q11 - Q16
- Allow about 2 hours and 45 minutes for this section

Section 1 (10 marks)

Attempt Questions 1-10

Allow about 15 minutes for this section

Question 1

The converse of “if it miaows it’s a cat” is

- (A) If it’s a cat it miaows. (B) If it’s not a cat it doesn’t miaow.
(C) If it doesn’t miaow it isn’t a cat. (D) If it’s a cat it doesn’t miaow.

Question 2

The contrapositive of “if it miaows it’s a cat” is

- (A) If it’s a cat it miaows. (B) If it’s not a cat it doesn’t miaow.
(C) If it doesn’t miaow it isn’t a cat. (D) If it’s a cat it doesn’t miaow.

Question 3

The straight line $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$ has symmetric (or Cartesian) equation

- (A) $x - 1 = y - 4 = \frac{z-3}{2}$ (B) $x - 1 = y - 4 = \frac{z+3}{2}$
(C) $1 - x = y - 4 = \frac{z-3}{2}$ (D) $1 - x = y - 4 = \frac{z+3}{2}$

Multiple choice continues on the following page

Multiple choice (continued)

Question 4

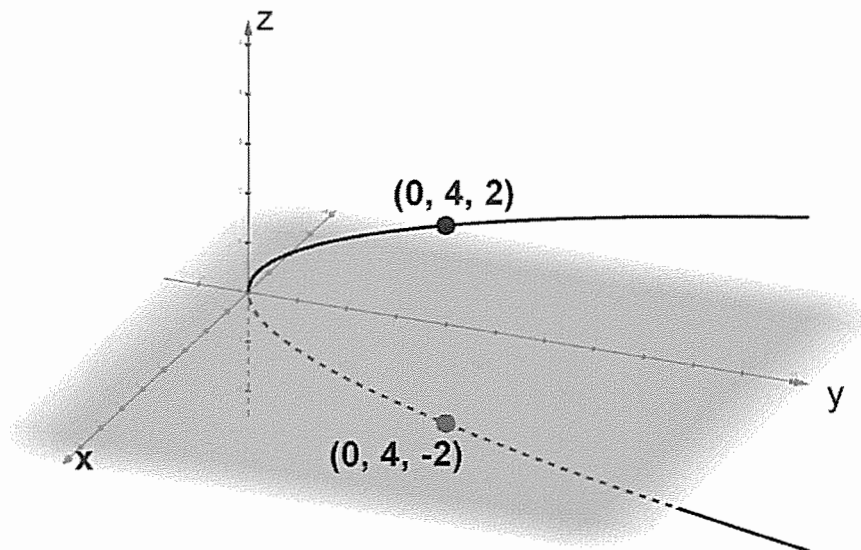
The equation of the curve in 3 dimensional space below is

(A) $x = t^2, y = t, z = 0$

(B) $x = 0, y = t^2, z = t$

(C) $x = 0, y = t, z = t^2$

(D) $x = t^2, y = t^2, z = 0$



Question 5

$$\int \frac{1}{\sqrt{6x - x^2}} \cdot dx =$$

(A) $\sin^{-1} \left(\frac{x}{\sqrt{6}} \right) + C$

(B) $\sin^{-1} \left(\frac{x}{6} \right) + C$

(C) $\sin^{-1} \left(\frac{x-3}{\sqrt{6}} \right) + C$

(D) $\sin^{-1} \left(\frac{x-3}{3} \right) + C$

Question 6

If $f(x)$ is even then $\int_{-a}^a f(x) \cdot dx =$

(A) $2 \int_0^a f(a-x) \cdot dx$

(B) $2 \int_0^{-a} f(a-x) \cdot dx$

(C) $2 \int_a^0 f(x) \cdot dx$

(D) All of the above.

Multiple choice continues on the following page

Multiple choice (continued)

Question 7

$z^2 + pz + q = 0, p, q$ real has one solution $z = 3 - 2i$. The values of p and q are:

(A) $p = 6, q = 13$

(B) $p = -6, q = 13$

(C) $p = -6, q = -13$

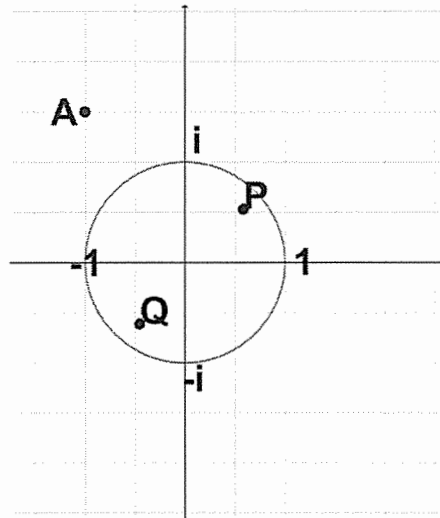
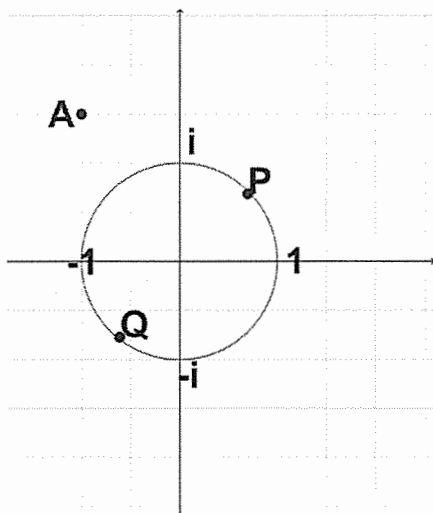
(D) $p = -6, q = -13$

Question 8

If the complex number z has $|z| > 1$ and $\frac{\pi}{2} < \text{Arg}(z) < \pi$ and $\overrightarrow{OA} = z$, then if the two square roots of z are $\overrightarrow{OP}$ and $\overrightarrow{OQ}$, P and Q could be at:

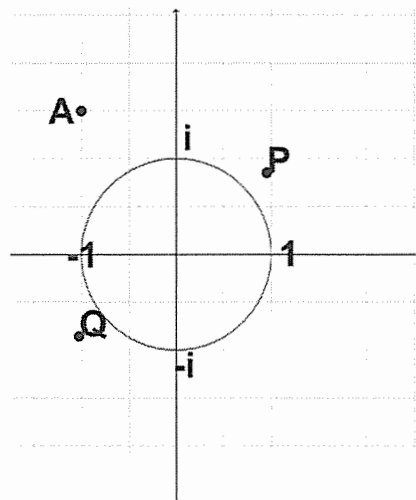
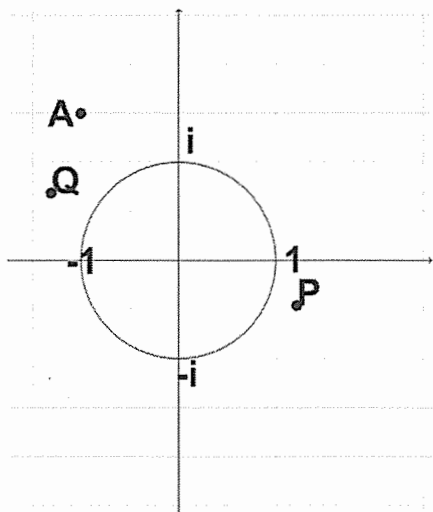
(A)

(B)



(C)

(D)



Multiple choice continues on the following page

Multiple choice (continued)

Question 9

The acceleration of a particle is given by $\ddot{x} = \sec^2 x$, where x is the displacement in metres. The rule for the velocity could be:

(A) $v = \sqrt{\tan x}$

(B) $v = \sqrt{2 \tan x}$

(C) $v = \tan x$

(D) $v = 2 \tan x$

Question 10

A particle moves with Simple Harmonic Motion (SHM) with equation $x = a \sin nt$. It first RETURNS to its starting point after $\frac{\pi}{3}$ seconds and is travelling at 15m/s forwards at this time.

The displacement of the particle of the particle is given by

(A) $x = \frac{5}{2} \sin 6t$

(B) $x = -\frac{5}{2} \sin 6t$

(C) $x = 5 \sin 3t$

(D) $x = -5 \sin 3t$

Examination continues on the following page

Section II (90 marks)

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Start the answers to each question on a separate page in your answer booklet.

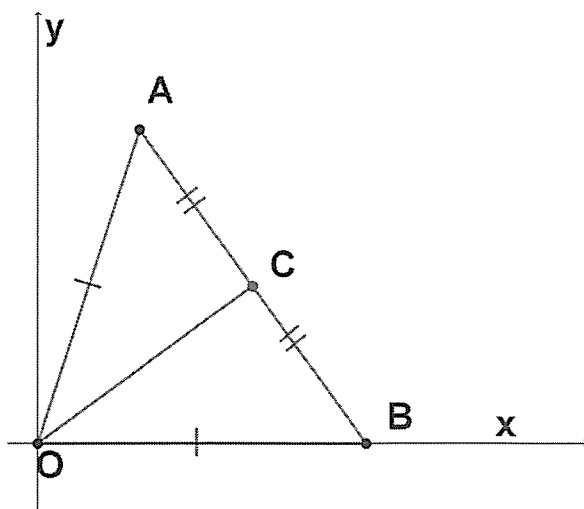
In Questions 11-16 your responses should include all relevant mathematical reasoning and/ or calculations.

Question 11 (15 marks)

Marks

- (a) (i) Find $\frac{-1+i\sqrt{3}}{1+i}$ in Cartesian form. 2
- (ii) Express $-1 + i\sqrt{3}$ and $1 + i$ in modulus/ argument form. 2
- (iii) Hence find the exact value of $\tan \frac{5\pi}{12}$. 2
- (b) (i) By letting $(x + iy)^2 = 7 + 24i$, x, y real, find $\sqrt{7 + 24i}$ in Cartesian form. 3
- (ii) Hence solve the equation $z^2 - 3iz - (4 + 6i) = 0$. 2
- (c) In the Argand diagram below, $\overrightarrow{OA} = z_1$, $\overrightarrow{OB} = z_2$, $OA = OB$ and $\overrightarrow{AC} = \overrightarrow{CB}$. 2

Express $\overrightarrow{OC}$ in terms of z_1 and z_2 .



- (d) Using DeMoivre's Theorem, find the 3 cube roots of $-4\sqrt{2} + 4i\sqrt{2}$. 2
- Leave your answers in modulus/argument form.

Examination continues on the following page

Question 12 (15 marks)**Marks**

- (a) Find $\int \sin^{-1}(x) \cdot dx$ 3
- (b) Find $\int \frac{1}{2+\cos x} \cdot dx$ 3
- (c) (i) Find A, B and C so that $\frac{A}{x-1} + \frac{Bx+C}{x^2+4} = \frac{-7x-18}{(x-1)(x^2+4)}$. 3
- (ii) Hence find $\int \frac{-7x-18}{(x-1)(x^2+4)} \cdot dx$ 2
- (d) (i) Using the substitution $x = \sec \theta$ or otherwise show by integrating that $\int \frac{1}{\sqrt{x^2-1}} \cdot dx = \ln(x + \sqrt{x^2-1})$ 2
- (ii) Find $\int \sqrt{\frac{x-1}{x+1}} \cdot dx$ 2

Question 13 (15 marks)

- (a) Prove if a is divisible by 5, $a^2 - 5a$ is divisible by 50. 4
(You may assume that the product of two consecutive numbers is even).
- (b) (i) Prove by contradiction that $\sqrt{2}$ is irrational. 2
- (ii) Prove by contraposition that if n^2 is not divisible by 18, n is not divisible by 6. 2
- (iii) Using (i), prove that $\sqrt{18}$ is irrational. 1
- (c) (i) Prove for all real a and b that $a^2 + b^2 \geq 2ab$. 1
- (ii) Hence prove for all $a \geq 0, a + \frac{1}{a} \geq 2$ 1
- (iii) Hence prove for all $a_1, a_2 \geq 0$,
 $(a_1 + a_2)(\frac{1}{a_1} + \frac{1}{a_2}) \geq 4$ 1
- (iv) Hence prove by induction for all $a_1, a_2, a_3 \dots a_n \geq 0$ 3
 $(a_1 + a_2 + a_3 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n} \right) \geq n^2$
(Note: you have already proven this for $n = 1$ and $n = 2$)

Examination continues on the following page

Question 14 (15 marks)**Marks**

(a) A particle is moving so that $v^2 = n^2(a^2 - x^2)$.

(i) Show that the particle is moving in Simple Harmonic Motion.

1

(ii) If the particle is moving at $\sqrt{15}m/s$ when it is $2m$ to the right of the centre of motion and at $2\sqrt{3}m/s$ when it is $4m$ to the right of the centre of motion, find the period and amplitude of the motion.

3

(b) A particle is moving so that $x = 2 \sin 3t - 2\sqrt{3} \cos 3t$.

(i) By expressing the motion of the particle in the form $A \sin(3t - \alpha)$, find the period and amplitude of the motion.

3

(ii) Find the initial position and velocity of the particle.

2

(c) A $20kg$ projectile is launched upwards from the ground at $500m/s$. It experiences force from gravity of 200 Newtons and air resistance in the opposite direction to its motion of $\frac{v^2}{18}$ Newtons.

Letting the ground be $x = 0$,

(i) Show that the weight's acceleration is given by $\ddot{x} = -10 - \frac{v^2}{360}$

1

(ii) Show that $x = 180 \ln \left(\frac{253600}{3600 + v^2} \right)$ and find the maximum height the projectile reaches.

3

(iii) Show that the time taken for the projectile to reach a velocity of $v m/s$ is given by $t = 6 \tan^{-1} \left(\frac{25}{3} \right) - 6 \tan^{-1} \frac{v}{60}$ and find the time taken for the projectile to reach its maximum height.

2

Examination continues on the following page

Question 15 (15 marks)**Marks**

- (a) (i) Show *by finding it* that the only point of intersection between

3

the straight line $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 10 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$ and the sphere

$(x - 5)^2 + (y + 2)^2 + (z - 3)^2 = 38$ is the point $\begin{pmatrix} 7 \\ 1 \\ 8 \end{pmatrix}$.

- (ii) Show *by finding it* that the point of intersection between the lines

3

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 10 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 15 \\ -1 \\ 6 \end{pmatrix} + \delta \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix}$ is also the point $\begin{pmatrix} 7 \\ 1 \\ 8 \end{pmatrix}$.

- (iii) Show that the line $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 15 \\ -1 \\ 6 \end{pmatrix} + \delta \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix}$ is also a tangent to

2

the sphere $(x - 5)^2 + (y + 2)^2 + (z - 3)^2 = 38$ by showing that it is perpendicular to the radius of the sphere at $\begin{pmatrix} 7 \\ 1 \\ 8 \end{pmatrix}$.

- (iv) Show that the straight line $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 11 \end{pmatrix} + \gamma \begin{pmatrix} 6 \\ 1 \\ -3 \end{pmatrix}$ is also a

3

tangent to the sphere $(x - 5)^2 + (y + 2)^2 + (z - 3)^2 = 38$ by finding

the perpendicular distance from the line $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 11 \end{pmatrix} + \gamma \begin{pmatrix} 6 \\ 1 \\ -3 \end{pmatrix}$

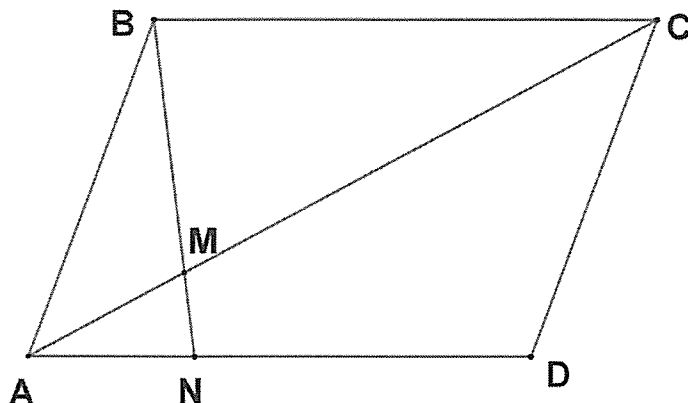
to the centre of the sphere.

Question 15 continues on the following page

Question 15 (continued)

Marks

- (b) ABCD is a parallelogram. N is on AD so that $\overrightarrow{AN} = \frac{1}{3} \overrightarrow{AD}$. M is on the diagonal AC. (See diagram).



- (i) Letting $\overrightarrow{AB} = \underline{p}$ and $\overrightarrow{AD} = \underline{q}$, express the vector $\overrightarrow{NB}$ in terms of $\underline{p}$ and $\underline{q}$. 1
- (ii) By letting $\overrightarrow{AM} = \lambda \overrightarrow{AC}$ and $\overrightarrow{MB} = \delta \overrightarrow{NB}$, prove that $\overrightarrow{AM} = \frac{1}{4} \overrightarrow{AC}$ 3

Question 16 (15 marks)

- (a) (i) Using Euler's theorem, show that if $z = e^{i\theta} = \cos \theta + i \sin \theta$ then $z^n + z^{-n} = 2 \cos n\theta$. 1
- (ii) Hence show that 2
- $$\cos^7 \theta = \frac{1}{64} (\cos 7\theta + 7 \cos 5\theta + 21 \cos 3\theta + 35 \cos \theta)$$
- (iii) Hence find $\int_0^{\frac{\pi}{2}} \cos^7 \theta \cdot d\theta$ 2
- (b) If $I_n = \int_0^{\frac{\pi}{2}} \cos^n \theta \cdot d\theta$
- (i) Show that $I_n = \frac{n-1}{n} I_{n-2}$ 2
- (ii) Hence find $\int_0^{\frac{\pi}{2}} \cos^7 \theta \cdot d\theta$ 2

Question 16 continues on the following page

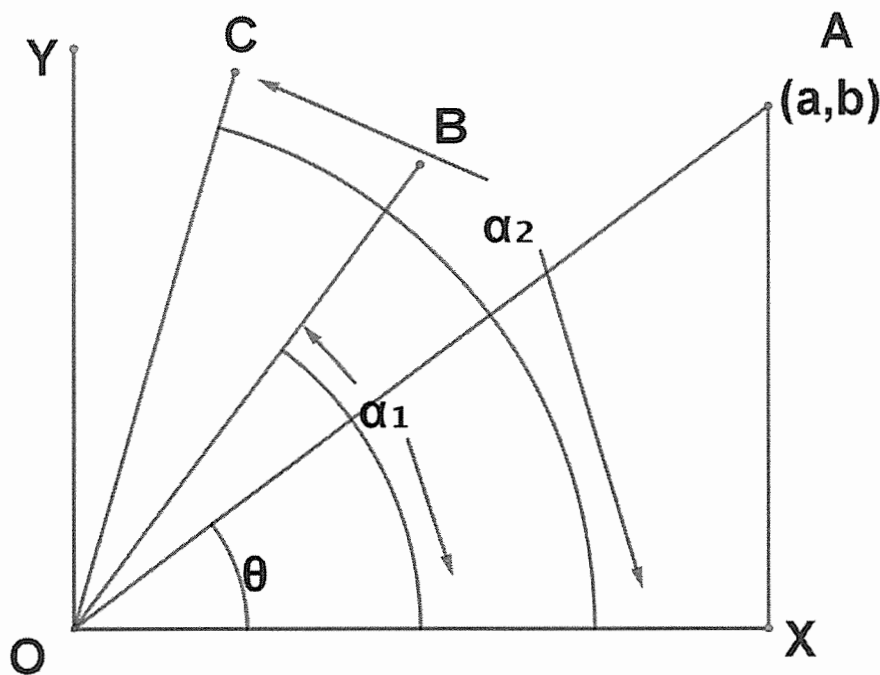
Question 16 (continued)

Marks

- (c) A projectile is launched from O at V m/s so that it will hit the point $A(a, b)$ on the slope. Air resistance is neglected and the acceleration due to gravity is g m/s².

OX is horizontal so that $\angle AOX = \theta$.

$\angle XOB = \alpha_1$ and $\angle XOC = \alpha_2$ so that α_1 and α_2 are the two angles with the horizontal the projectile can be fired at so that it hits the target at A . (see diagram).



- (i) Given that $y = \frac{-gx^2}{2V^2} (1 + \tan^2 \alpha) + x \tan \alpha$, and given

2

(as shown in the diagram) that $\alpha_1 < \alpha_2$, show that

$$\tan \alpha_1 = \frac{V^2 - \sqrt{V^4 - g^2 a^2 - 2V^2 gb}}{ga} \text{ and}$$

$$\tan \alpha_2 = \frac{V^2 + \sqrt{V^4 - g^2 a^2 - 2V^2 gb}}{ga}$$

- (ii) By showing that $\tan (\alpha_1 + \alpha_2) = \tan (90^\circ + \theta)$, show that

4

$$\angle COY = \angle BOA.$$

END OF EXAMINATION!!!



GIRRAWEE HIGH SCHOOL

MATHEMATICS EXTENSION 2

2022 TRIAL HIGHER SCHOOL CERTIFICATE

Student Number: Solutions!

This Booklet contains the answer sheet for Section 1 and Writing Booklet for Section 2.

Section 1 ANSWER SHEET

Select the alternative A, B, C or D that best answers the question.

1.	A	☒	B	☐	C	☐	D	☐
2.	A	☐	B	☒	C	☐	D	☐
3.	A	☐	B	☐	C	☒	D	☐
4.	A	☐	B	☒	C	☐	D	☐
5.	A	☐	B	☐	C	☐	D	☒
6.	A	☒	B	☐	C	☐	D	☐
7.	A	☐	B	☒	C	☐	D	☐
8.	A	☐	B	☐	C	☐	D	☒
9.	A	☐	B	☒	C	☐	D	☐
10.	A	☐	B	☐	C	☐	D	☒

Instructions

- If you need more paper for Section 2, please ask your supervisor.
- Write your student number on every booklet you use.
- Write on both sides of each sheet of paper.

Total number of booklets used _____.

GHS 4U Trial 2022 Solutions: p.1

(1) (A) (2) B (3) C (4) B (5) D (6) A (7) B
(8) D (9) B (10) D

(1) (A) If it's a cat it miaows. $\rightarrow$ From If A then B to If B then A.
(2) (B) If it isn't a cat it doesn't miaow. $\rightarrow$ From if A then B to If not B then not A.

$$(3) x = 1 - \lambda \quad y = 4 + \lambda \quad z = 3 + 2\lambda$$

$$\therefore \lambda = 1 - x \quad \therefore \lambda = y - 4 \quad \therefore \lambda = \frac{z - 3}{2}$$

$$\therefore 1 - x = y - 4 = \frac{z - 3}{2} \quad (C)$$

(4) B $[x=0, y=t^2, z=t]$.

$$(5) \int \frac{1}{\sqrt{6x-x^2}} dx$$

$$= \int \frac{1}{\sqrt{9-(x-3)^2}} dx$$

$$= \sin^{-1}\left(\frac{x-3}{3}\right) + C. \quad (D)$$

$$(6) \int_{-a}^a f(x) dx$$

$$= 2 \int_0^a f(x) dx, \quad f(x) \text{ even}$$

$$= 2 \int_0^a f(a-x) dx \quad (A)$$

(7) $\alpha = 3 - 2i$ solution
 $\beta = 3 + 2i$ solution
 (conjugate root theorem).
 $\alpha + \beta = -p = 6$
 $p = -6$
 $\alpha\beta = q = 13 \quad (B)$
 $p = -6, q = 13$

(8) One root in $\mathbb{Q}_1$.
 Other in $\mathbb{Q}_3$
 $|\sqrt{2}| > 1$ as $|z| > 1$.
 (D)

(9) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \sec^2 x$
 $\frac{d}{dx} (v^2) = 2 \sec^2 x$
 $v^2 = \int 2 \sec^2 x dx$
 $= 2 \tan x + C$
 $v = \sqrt{2 \tan x} \quad (B)$

(10) Period = $\frac{2\pi}{3} \Rightarrow k=3$.
 $x = a \sin 3t$
 $v = 3a \cos 3t$
 $t = \frac{\pi}{3}, v = 15, x = -5 \sin 3t \quad (D)$
 $3a \cos \pi = 5$
 $a = -5$

$$\begin{aligned} Q. (11)(a)(i) \quad \frac{-1+i\sqrt{3}}{1+i} &\times \frac{(1-i)}{(1-i)} \\ &= \frac{(\sqrt{3}-1)}{2} + i \frac{(\sqrt{3}+1)}{2} \end{aligned}$$

$$\begin{aligned} (ii) \quad -1+i\sqrt{3} &= 2 \operatorname{cis} \frac{2\pi}{3} \\ 1+i &= \sqrt{2} \operatorname{cis} \frac{\pi}{4} \end{aligned}$$

$$(iii) \quad \frac{-1+i\sqrt{3}}{1+i} = \frac{2 \operatorname{cis} \frac{2\pi}{3}}{\sqrt{2} \operatorname{cis} \frac{\pi}{4}}$$

$$\therefore \frac{\sqrt{3}-1}{2} + i \frac{(\sqrt{3}+1)}{2} = \sqrt{2} \operatorname{cis} \frac{5\pi}{12}$$

Equating parts,

$$\frac{\sqrt{3}+1}{2} = \sqrt{2} \sin \frac{5\pi}{12} \quad (1)$$

$$\& \frac{\sqrt{3}-1}{2} = \sqrt{2} \cos \frac{5\pi}{12} \quad (2)$$

$$\begin{aligned} (1) \div (2) + \tan \frac{5\pi}{12} &= \frac{\sqrt{3}+1}{\sqrt{3}-1} \\ &= 2+\sqrt{3} \end{aligned}$$

$$(b)(i) \quad (x+iy)^2 = 7+24i$$

$$(x^2-y^2)+2ixy = 7+24i$$

Equating reals

$$x^2-y^2 = 7 \quad (1)$$

Equating imaginaries

$$2xy = 24$$

$$y = \frac{12}{x} \quad (2)$$

Sub. (2) in (1):

$$x^2 - \frac{144}{x^2} = 7$$

$$x^4 - 7x^2 - 144 = 0$$

$$(x^2-16)(x^2+9) = 0$$

As $x \in \mathbb{R}$, $x = \pm 4$.

Sub. $x=4$ in (2) | Sub. $x=-4$ in (2)

$$y = \frac{12}{4}$$

$$= 3$$

$$y = \frac{12}{-4}$$

$$= -3$$

$$\sqrt{7+24i} = \pm (4+3i)$$

$$(ii) \quad z^2 - 3iz - (4+6i) = 0$$

$$z = \frac{3i \pm \sqrt{(-3i)^2 - 4 \times 1 \times -(4+6i)}}{2}$$

$$= \frac{3i \pm \sqrt{7+24i}}{2}$$

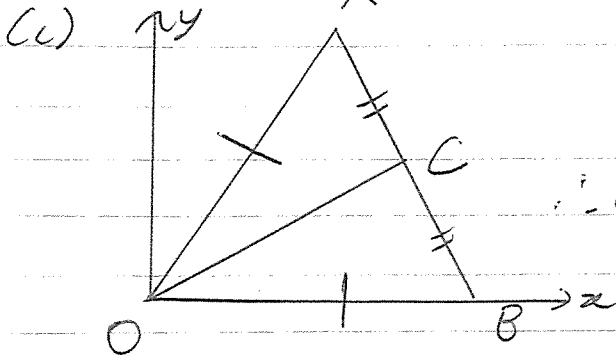
$$= \frac{3i + (4+3i)}{2} \text{ or } \frac{3i - (4+3i)}{2}$$

[using (i)]

$$\# z = 2+3i \text{ or } z = -2$$

p. 3

Q. (11) (continued)



$$(i) \vec{AB} = \underline{b} - \underline{a}$$

$$\therefore \vec{AC} = \frac{1}{2}(\underline{b} - \underline{a})$$

$$\therefore \vec{OC} = \vec{OA} + \vec{AC}$$

$$= \underline{a} + \frac{1}{2}(\underline{b} - \underline{a})$$

$$= \underline{\underline{\frac{1}{2}(\underline{a} + \underline{b})}}$$

(d) $-4\sqrt{2} + 4i\sqrt{2}$

$$= 8 \operatorname{cis} \frac{3\pi}{4}$$

$$\therefore \sqrt[3]{-4\sqrt{2} + 4i\sqrt{2}}$$

$$= 2 \operatorname{cis} \frac{\pi}{4}$$

$$= 2 \operatorname{cis} \frac{11\pi}{12}$$

$$= 2 \operatorname{cis} -\frac{5\pi}{12}$$

p. 4

$$Q. (12) (a) \int \sin^{-1}(x) \cdot dx$$

$$u = \sin^{-1}(x) \quad v = x$$

$$u' = \frac{1}{\sqrt{1-x^2}} \quad v' = 1$$

$$\text{By } \int u v' \cdot dx = uv - \int v u' \cdot dx$$

$$\int \sin^{-1}(x) \cdot dx = x \sin^{-1}(x) - \int \frac{x}{\sqrt{1-x^2}} \cdot dx$$

$$= x \sin^{-1}(x) + \frac{1}{2} \int \frac{-2x}{\sqrt{1-x^2}} \cdot dx$$

$$= x \sin^{-1}(x) + \frac{1}{2} \times 2 \sqrt{1-x^2} + C \quad \left[\text{By } \int f'(x) \cdot [f(x)]^n \cdot dx \right.$$

$$\left. = \frac{1}{n+1} [f(x)]^{n+1} \right.$$

$$= x \sin^{-1}(x) + \sqrt{1-x^2} + C.$$

$$(b) \int \frac{1}{2+\cos x} \cdot dx \quad t = \tan\left(\frac{x}{2}\right) \quad \therefore dx = \frac{2}{1+t^2} \cdot dt$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2\left(\frac{x}{2}\right)$$

$$= \frac{1+t^2}{2}$$

$$= \int \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} \cdot dt$$

$$= \int \frac{2}{3+t^2} \cdot dt$$

$$= \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{t}{\sqrt{3}}\right) + C$$

$$= \frac{2}{\sqrt{3}} \tan^{-1}\left[\frac{\tan\left(\frac{x}{2}\right)}{\sqrt{3}}\right] + C.$$

$$(c) (i) \frac{A}{x-1} + \frac{Bx+C}{x^2+4} = -7x-18$$

$$\therefore A(x^2+4) + (Bx+C)(x-1) = -7x-18.$$

$$\text{Sub. in } x=1.$$

$$5A$$

$$= -25 \Rightarrow A = -5.$$

$$\text{Sub. in } x=0, A=-5.$$

$$-20 - C = -18 \Rightarrow C = -2.$$

$$\text{Sub. in } x=2, A=-5, C=-2.$$

$$-40 + (2B-2) = -32.$$

$$2B-2 = 8$$

$$B = 5.$$

$$\underline{A = -5, B = 5, C = -2}$$

(ii) Hence

$$\int \frac{-7x-18}{(x-1)(x^2+4)} \cdot dx$$

$$= \int \frac{-5}{x-1} + \frac{5x}{x^2+4} - \frac{2}{x^2+4} \cdot dx$$

$$= -5 \ln(x-1) + 5 \ln(x^2+4)$$

$$- \tan^{-1}\left(\frac{x}{2}\right) + C.$$

$$Q. (12) (d) (i) \int \frac{1}{\sqrt{x^2-1}} dx$$

$$x = \sec \theta.$$

$$dx = \sec \theta \tan \theta \cdot d\theta$$

$$= \int \frac{1}{\tan \theta} \cdot \sec \theta \tan \theta \cdot d\theta$$

$$= \int \sec \theta \cdot d\theta$$

$$= \int \frac{\sec^2 \theta + \sec \theta \tan \theta}{\sec \theta + \tan \theta} \cdot d\theta$$

$$= \ln [\sec \theta + \tan \theta] + C$$

$$= \underline{\underline{\ln [x + \sqrt{x^2-1}] + C.}}$$

$$(ii) \int \sqrt{\frac{x-1}{x+1}} \cdot dx \quad \begin{matrix} \times \sqrt{x-1} \\ \div \sqrt{x-1} \end{matrix}$$

$$= \int \frac{x-1}{\sqrt{x^2-1}} \cdot dx$$

$$= \int \frac{x}{\sqrt{x^2-1}} \cdot dx - \int \frac{1}{\sqrt{x^2-1}} \cdot dx$$

$$= \frac{1}{2} \int \frac{2x}{\sqrt{x^2-1}} \cdot dx - \int \frac{1}{\sqrt{x^2-1}} \cdot dx$$

$$= \underline{\underline{\sqrt{x^2-1} - \ln [x + \sqrt{x^2-1}] + C.}}$$

Q. (13)(a) Either: Let a be divisible by 5 i.e. $a = 5k$.

$$\begin{aligned} a^2 - 5a &= [5k]^2 - 5 \times 5k \\ &= 25k^2 - 25k \\ &= 25k(k-1) \end{aligned}$$

As the product of 2 consecutive numbers is even:

$$\text{Let } k(k-1) = 2m$$

$$\text{So } a^2 - 5a = 25 \times 2m$$

$$= 50m$$

which is divisible by 5 (OR)

Let a be divisible by 5

Case 1 but NOT by 10

$$\therefore a = 10k-5, k \in \mathbb{Z}^+$$

$$\begin{aligned} a^2 - 5a &= (10k-5)^2 - 5(10k-5) \\ &= (10k-5)(10k-5-5) \\ &= 5(2k-1) \times 10(k-1) \\ &= 50(k-1)(2k-1) \end{aligned}$$

which is divisible by 50.

Case 2: a divisible by 10;

$$\begin{aligned} a &= 10k. \\ a^2 - 5a &= (10k)^2 - 5(10k) \\ &= 100k^2 - 50k \\ &= 50k(2k-1) \end{aligned}$$

which is divisible by 50.

(b)(i) Let $\sqrt{2}$ be rational

$$\text{i.e. } \sqrt{2} = \frac{p}{q}, p, q \in \mathbb{Z}^+, p, q \text{ mutually prime.}$$

Squaring BS:

$$2 = \frac{p^2}{q^2}$$

$$2q^2 = p^2 \quad (1)$$

$\therefore$ As q NOT factor of p

$2q^2$ is a factor of p

$$p = 2k$$

$$p^2 = 4k^2 \quad (2)$$

Sub. (1) in (2):

$$2q^2 = 4k^2$$

$$q^2 = 2k^2$$

$\Rightarrow 2$ is a factor of q^2

$\therefore 2$ is a factor of q .

BUT p & q are mutually prime.

Contradiction.

$\sqrt{2}$ is IRRATIONAL.

(ii) Contrapositive.

If n is divisible by 6,
 n^2 is divisible by 18.

$$\text{Let } n = 6k.$$

$$n^2 = 36k^2$$

$$= 18 \times 2k^2$$

which is divisible by 18.

$$(iii) \sqrt{18} = 3\sqrt{2}.$$

If $\sqrt{18}$ is rational,

$$\sqrt{18} = \frac{p}{q}$$

$$\therefore \sqrt{2} = \frac{p}{3q} \text{ which is rational.}$$

But $\sqrt{2}$ is NOT rational.

$\therefore \sqrt{18}$ is irrational.

p-7

Q.13) (i) (ii) If a, b real

$$\begin{aligned} (a-b)^2 &\geq 0 \\ a^2 + b^2 &\geq 2ab \quad +2ab \text{ by } \text{BS.} \end{aligned}$$

(ii) Either using (i)

$$a^2 + 1 \geq 2a$$

$$a + \frac{1}{a} \geq 2 \quad \text{by BS by } a, a > 0$$

$$\text{or } \left(\sqrt{a} - \frac{1}{\sqrt{a}}\right)^2 \geq 0$$

$$a - 2 + \frac{1}{a} \geq 0$$

$$a + \frac{1}{a} \geq 2.$$

$$(iii) (a_1 + a_2) \left(\frac{1}{a_1} + \frac{1}{a_2} \right)$$

$$= 1 + \frac{a_1}{a_2} + \frac{a_2}{a_1} + 1$$

$$\geq 1 + 2 + 1$$

$$= 4 \quad \left[\text{as } \frac{a_1}{a_2} + \frac{a_2}{a_1} \geq 2 \text{ by (ii)} \right].$$

(iv) Assume true for $n=k$

$$\text{i.e. } (a_1 + a_2 + \dots + a_k) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k} \right) \geq k^2$$

Prove true for $n=k+1$

$$\text{i.e. RTP } (a_1 + a_2 + \dots + a_k + a_{k+1}) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k} + \frac{1}{a_{k+1}} \right) \geq (k+1)^2$$

LHS

$$\begin{aligned} & \left[(a_1 + a_2 + \dots + a_k) + a_{k+1} \right] \left[\left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k} \right) + \frac{1}{a_{k+1}} \right] \\ &= (a_1 + a_2 + \dots + a_k) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k} \right) + \left(\frac{a_1}{a_{k+1}} + \frac{a_2}{a_{k+1}} + \dots + \frac{a_k}{a_{k+1}} \right) + \left(\frac{a_{k+1}}{a_1} + \dots + \frac{a_{k+1}}{a_k} \right) \\ & \quad + 1 \end{aligned}$$

$$= (a_1 + a_2 + \dots + a_k) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k} \right) + \left[\frac{a_1}{a_{k+1}} + \frac{a_{k+1}}{a_1} + \dots + \frac{a_k}{a_{k+1}} + \frac{a_{k+1}}{a_k} \right] + 1$$

$$\geq k^2 \quad \text{[By assumption]} \quad + \left[2 + 2 + \dots + 2 \right] + 1 \quad \text{[using (ii)]}$$

$$= k^2 \quad + 2k \quad + 1$$

$$= (k+1)^2$$

= RHS QED. True by the principle of mathematical induction.

Q. (14) (a) (i) $v^2 = n^2(a^2 - x^2)$ p. 8
 $= n^2 a^2 - n^2 x^2$

$$\frac{d}{dx}(v^2) = -2n^2 x$$

$$\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right) = -n^2 x$$

Particle is moving with SHM.

(ii) $v^2 = n^2(a^2 - x^2)$

$$(\sqrt{15})^2 = n^2(a^2 - 2^2) \quad (2\sqrt{3})^2 = n^2(a^2 - 4^2)$$

$$15 = n^2 a^2 - 4n^2(1) \quad 12 = n^2 a^2 - 16n^2(2)$$

$$\therefore (1) - (2): 12n^2 = 3$$

$$n^2 = \frac{1}{4}$$

$$n = \frac{1}{2}$$

$$\text{Period} = 4\pi$$

Sub. $n = \frac{1}{2}$ in (1):

$$15 = \frac{1}{4}(a^2 - 4)$$

$$8 = a$$

$$\text{Amplitude} = 8$$

(b) (i) $2\sin 3t - 2\sqrt{3}\cos 3t = A\sin(3t - \alpha)$

$$2\sin 3t - 2\sqrt{3}\cos 3t = A\sin 3t \cos \alpha - A\cos 3t \sin \alpha$$

Equating parts

$$2\sin 3t = A\sin 3t \cos \alpha \quad 2\sqrt{3}\cos 3t = A\cos 3t \sin \alpha$$

$$2 = A\cos \alpha \quad (1) \quad 2\sqrt{3} = A\sin \alpha$$

$$(1)^2 + (2)^2 = A^2[\cos^2 \alpha + \sin^2 \alpha] = 2^2 + (2\sqrt{3})^2$$

Sub. $A = 4$ in (1):

$$2 = 4\cos \alpha \Rightarrow \cos \alpha = \frac{1}{2}$$

$$\alpha = \frac{\pi}{3}$$

$$A = 4$$

Sub. $A = 4$ in (2):

$$2\sqrt{3} = 4\sin \alpha$$

$$\frac{\sqrt{3}}{2} = \sin \alpha$$

$$x = 4\sin\left(3t - \frac{\pi}{3}\right) \quad \text{Period} = \frac{2\pi}{3} \quad \text{Amplitude} = 4$$

(ii) Initial position = $4\sin\left(-\frac{\pi}{3}\right)$ Initial velocity: $v = 12\cos\left(3t - \frac{\pi}{3}\right)$

$$= -2\sqrt{3} \text{ m}$$

$$2\sqrt{3} \text{ m LEFT of O.}$$

$$t=0: v = 12\cos\left(-\frac{\pi}{3}\right) = \underline{6 \text{ m/s.}}$$

p.9

Q. (14) (xi) $\downarrow \quad \downarrow$
 $200\text{N} \left(\frac{1}{18} \text{N} \right)$

$$F = ma = -200 - \frac{v^2}{18}$$

$$20a = -200 - \frac{v^2}{18}$$

$$a = -10 - \frac{v^2}{360}$$

$$\begin{aligned} \text{(ii)} \quad v \cdot \frac{dv}{dx} &= -10 - \frac{v^2}{360} \\ &= -\frac{3600 + v^2}{360} \end{aligned}$$

$$\frac{dv}{dx} = -\frac{3600 + v^2}{360v}$$

$$\frac{dx}{dv} = -\frac{360v}{3600 + v^2}$$

$$\begin{aligned} x &= -180 \int \frac{2v}{3600 + v^2} \cdot dv \\ &= -180 \ln(3600 + v^2) + C \end{aligned}$$

$$x = 0, v = 500$$

$$0 = -180 \ln(3600 + 500^2) + C$$

$$180 \ln(253600) = C$$

$$x = 180 \ln \left[\frac{253600}{3600 + v^2} \right]$$

$$\text{Max. height: } x \text{ when } v = 0$$

$$= 180 \ln \left[\frac{253600}{3600 + 0^2} \right]$$

$$= 765.86 \dots$$

$$\approx 766 \text{m (nearest m)}$$

$$\text{(iii)} \quad \frac{dv}{dt} = \frac{-3600 - v^2}{360}$$

$$\therefore \frac{dt}{dv} = \frac{-360}{3600 + v^2}$$

$$t = \int \frac{-360}{3600 + v^2} \cdot dv$$

$$= -360 \times \frac{1}{60} \tan^{-1} \left(\frac{v}{60} \right) + C$$

$$t = -6 \tan^{-1} \left(\frac{v}{60} \right) + C$$

$$t = 0, v = 500$$

$$0 = -6 \tan^{-1} \left(\frac{500}{60} \right) + C$$

$$= -6 \tan^{-1} \left(\frac{25}{3} \right) + C$$

$$6 \tan^{-1} \left(\frac{25}{3} \right) = C$$

$$t = 6 \tan^{-1} \left(\frac{25}{3} \right) - 6 \tan^{-1} \left(\frac{v}{60} \right)$$

Time for max height is

t when $v = 0$:

$$= 6 \tan^{-1} \left(\frac{25}{3} \right)$$

$$= 8.708 \dots$$

Time to max. height $\approx 8.7 \text{ seconds}$

p.10

Q.15(a) (i) $x = 1 + 2\lambda$

$y = 10 - 3\lambda$

$z = 5 + \lambda$

Note: Could also use

$\Delta = (-6)^2 - 4 \times 1 \times 9 = 0$
to show tangent.

$(x-5)^2 + (y+2)^2 + (z-3)^2 = 38$

$(2\lambda-4)^2 + (12-3\lambda)^2 + (2+\lambda)^2 = 38$

$14\lambda^2 - 84\lambda + 126 = 0$

$\lambda^2 - 6\lambda + 9 = 0$

$(\lambda-3)^2 = 0$

$\Rightarrow$ Only 1 point of intersection: $\lambda = 3$.

$x = 1 + 2 \times 3 = 7$

$y = 10 - 3 \times 3 = 1$

$z = 5 + 3 = 8$

Point of intersection = $\begin{pmatrix} 7 \\ 1 \\ 8 \end{pmatrix}$

(ii) Point of intersection between

$\begin{pmatrix} 1 \\ 10 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$ & $\begin{pmatrix} 15 \\ -1 \\ 6 \end{pmatrix} + \delta \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix}$

$x: 1 + 2\lambda = 15 + 4\delta \Rightarrow \lambda - 2\delta = 7 \quad (1)$

$y: 10 - 3\lambda = -1 - \delta \Rightarrow 3\lambda - \delta = 11 \quad (2)$

$z: 5 + \lambda = 6 - \delta \Rightarrow \lambda + \delta = 1 \quad (3)$

$4\lambda = 12$

$\lambda = 3$

Sub. $\lambda = 3$ in (3): $3 + \delta = 1$

$\delta = -2$

Sub. $\lambda = 3, \delta = -2$ in (1) to make sure it works:

$3 - 2 \times -2 = 7 \checkmark$

Point of intersection = $\begin{pmatrix} 1 \\ 10 \\ 5 \end{pmatrix} + 3 \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 15 \\ -1 \\ 6 \end{pmatrix} - 2 \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix}$

= $\begin{pmatrix} 7 \\ 1 \\ 8 \end{pmatrix}$

= $\begin{pmatrix} 7 \\ 1 \\ 8 \end{pmatrix} \checkmark$

PTD

(15)(a)(iii) Radius of sphere at $\begin{pmatrix} 7 \\ 8 \end{pmatrix}$

$$= \begin{pmatrix} 7 \\ 8 \end{pmatrix} - \begin{pmatrix} 5 \\ -2 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$$

Showing $\begin{pmatrix} 15 \\ -1 \\ 6 \end{pmatrix} + 5 \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix} \perp$ radius.

$$\begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$$

$$= 8 - 3 - 5$$

$$= 0.$$

As direction vector, radius = 0.

Line $\perp$ radius..

(iv) $\perp$ distance from $\begin{pmatrix} 1 \\ 0 \\ 11 \end{pmatrix} + \angle \begin{pmatrix} 6 \\ 1 \\ -3 \end{pmatrix} + \begin{pmatrix} 5 \\ -2 \\ 3 \end{pmatrix}$.

Diagram showing points A $\begin{pmatrix} 1 \\ 0 \\ 11 \end{pmatrix}$, B $\begin{pmatrix} 6 \\ 1 \\ -3 \end{pmatrix}$, and H $\begin{pmatrix} 5 \\ -2 \\ 3 \end{pmatrix}$. A line segment AB is drawn, and a perpendicular line segment AH is drawn from A to the line containing B. The vector $\underline{a} = \begin{pmatrix} 4 \\ -2 \\ -8 \end{pmatrix}$ is shown along AB, and the vector $\underline{b} = \begin{pmatrix} 6 \\ 1 \\ -3 \end{pmatrix}$ is shown along BH. The projection of $\underline{a}$ onto $\underline{b}$ is calculated as:

$$\text{Proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|^2} \underline{b}$$

$$= \frac{46}{46} \begin{pmatrix} 6 \\ 1 \\ -3 \end{pmatrix} = \begin{pmatrix} 6 \\ 1 \\ -3 \end{pmatrix}$$

Then

$$\vec{HB} = \vec{HA} + \vec{AB}$$

$$= \vec{AB} - \vec{AH}$$

$$= \begin{pmatrix} 6 \\ 1 \\ -3 \end{pmatrix} - \begin{pmatrix} 4 \\ -2 \\ -8 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$$

& $HB = |\vec{HB}| = \sqrt{2^2 + 3^2 + 5^2}$

$$= \sqrt{38} \text{ units}$$

[Can also use Pythagoras]

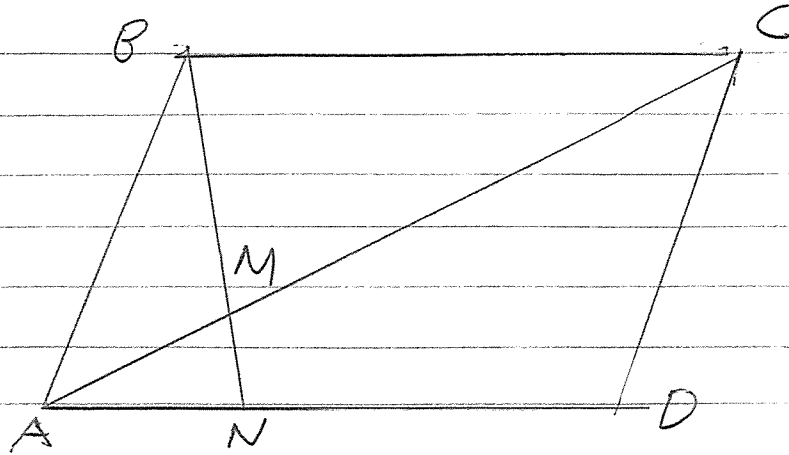
$$AH = \sqrt{4^2 + (-2)^2 + 8^2} = \sqrt{84}$$

$$AB = \sqrt{6^2 + 1^2 + (-3)^2} = \sqrt{46}$$

$$HB = \sqrt{(\sqrt{84})^2 - (\sqrt{46})^2}$$

$$= \sqrt{38} \text{ units}$$

(15)(b)



$$(i) \underline{AN} = \frac{1}{3} \underline{q}$$

$$\therefore \underline{NA} = -\frac{1}{3} \underline{q}$$

$$\begin{aligned} \underline{NB} &= \underline{NA} + \underline{AB} \\ &= -\frac{1}{3} \underline{q} + \underline{r} \\ &= \underline{r} - \frac{1}{3} \underline{q} \end{aligned}$$

$$(ii) \underline{AM} = \lambda \underline{AC}$$

$$= \lambda (\underline{r} + \underline{q})$$

$$\begin{aligned} \underline{MB} &= \delta \underline{NB} \\ &= \delta (\underline{r} - \frac{1}{3} \underline{q}) \end{aligned}$$

$$\underline{AB} = \underline{AM} + \underline{MB}$$

$$\underline{r} + 0 \underline{q} = \lambda \underline{r} + \lambda \underline{q} + \delta \underline{r} - \frac{1}{3} \delta \underline{q}$$

$$\underline{r} + 0 \underline{q} = (\lambda + \delta) \underline{r} + (\lambda - \frac{1}{3} \delta) \underline{q}$$

$$\text{Equating } \underline{r}: \lambda + \delta = 1 \quad (1)$$

$$\text{Equating } \underline{q}: \lambda - \frac{1}{3} \delta = 0 \quad (2)$$

$$\begin{aligned} \frac{4}{3} \delta &= 1 \\ \delta &= \frac{3}{4} \end{aligned}$$

$$\text{As } \lambda + \delta = 1, \lambda = \frac{1}{4}$$

$$\therefore \underline{AM} = \frac{1}{4} \underline{AC}$$

Q. p. 13

Q. (16) (a) (i) Euler's theorem

$$z^n = e^{ni\theta} = \cos n\theta + i \sin n\theta.$$

$$z^{-n} = e^{-ni\theta} = \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos n\theta - i \sin n\theta \quad (\cos \text{ even, } \sin \text{ odd})$$

$$\therefore z^n + z^{-n} = 2 \cos n\theta.$$

$$(ii) (z + z^{-1})^7 = z^7 + 7z^5 + 21z^3 + 35z + \frac{35}{z} + \frac{21}{z^3} + \frac{7}{z^5} + \frac{1}{z^7}$$

$$\left(z + \frac{1}{z}\right)^7 = \left(z^7 + \frac{1}{z^7}\right) + 7\left(z^5 + \frac{1}{z^5}\right) + 21\left(z^3 + \frac{1}{z^3}\right) + 35\left(z + \frac{1}{z}\right)$$

$$\therefore (2 \cos \theta)^7 = 2 \cos 7\theta + 14 \cos 5\theta + 42 \cos 3\theta + 70 \cos \theta.$$

$$128 \cos^7 \theta = 2 \cos 7\theta + 14 \cos 5\theta + 42 \cos 3\theta + 70 \cos \theta$$

$$\cos^7 \theta = \frac{1}{64} [\cos 7\theta + 7 \cos 5\theta + 21 \cos 3\theta + 35 \cos \theta.]$$

as required.

$$(iii) \int_0^{\frac{\pi}{2}} \cos^7 \theta \cdot d\theta = \frac{1}{64} \int_0^{\frac{\pi}{2}} \cos 7\theta + 7 \cos 5\theta + 21 \cos 3\theta + 35 \cos \theta \cdot d\theta$$

$$= \frac{1}{64} \left[\frac{1}{7} \sin 7\theta + \frac{7}{5} \sin 5\theta + 7 \sin 3\theta + 35 \sin \theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{64} \left[-\frac{1}{7} + \frac{7}{5} - 7 + 35 \right]$$

$$= \frac{1}{64} \times \frac{1024}{35} = \boxed{\frac{16}{35}}$$

$$(b) (i) I_n = \int_0^{\frac{\pi}{2}} \cos^n \theta \cdot d\theta \quad \begin{array}{l} u = \cos^{n-1} \theta \quad v = \sin \theta \\ u' = (n-1) \cos^{n-2} \theta \cdot -\sin \theta \quad v' = \cos \theta \end{array}$$

$$\text{By } \int u v' \cdot dx = uv - \int v u' \cdot dx$$

$$\int_0^{\frac{\pi}{2}} \cos^n \theta \cdot d\theta = [\cos^{n-1} \theta \sin \theta]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} \theta \cdot \sin^2 \theta \cdot d\theta$$

$$I_n = 0 + (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} \theta (1 - \cos^2 \theta) d\theta$$

$$I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$n I_n = (n-1) I_{n-2} \Rightarrow I_n = \frac{(n-1)}{n} I_{n-2}$$

p.14

$$Q.(16)(b)(ii) \quad I_1 = \int_0^{\frac{\pi}{2}} \cos \theta \cdot d\theta$$

$$= \left[\sin \theta \right]_0^{\frac{\pi}{2}}$$

$$= 1.$$

$$I_7 = \frac{6}{7} \times \frac{4}{5} \times \frac{2}{3} \times 1 \cdot I_1$$

$$= \frac{16}{35}$$

p. 15

Q. (16)(i)(i) Hits target at (a, b).

$$y = -\frac{gx^2}{2V^2} (1 + \tan^2 \alpha) + x \tan \alpha$$

$$b = -\frac{ga^2}{2V^2} - \frac{ga^2}{2V^2} \tan^2 \alpha + a \tan \alpha.$$

$$\times 2V^2 \quad \& + 2V^2 b.$$

$$0 = ga^2 \tan^2 \alpha - 2Va \tan \alpha + (ga^2 + 2V^2 b).$$

$$\therefore \tan \alpha = \frac{2V^2 a \pm \sqrt{4V^4 a^2 - 4ga^2 (ga^2 + 2V^2 b)}}{2ga^2}$$

$$= \frac{2V^2 a \pm \sqrt{4a^2 [V^4 - g^2 a^2 - 2gV^2 b]}}{2ga^2}$$

$$= \frac{2V^2 a \pm 2a \sqrt{V^4 - g^2 a^2 - 2gV^2 b}}{2ga^2} \quad \begin{matrix} \div 2a \\ \div 2a \end{matrix}$$

$$\tan \alpha = \frac{V^2 \pm \sqrt{V^4 - g^2 a^2 - 2gV^2 b}}{ga}.$$

$$\text{As } \alpha_2 > \alpha_1, \tan \alpha_2 = \frac{V^2 + \sqrt{V^4 - g^2 a^2 - 2gV^2 b}}{ga}$$

$$\& \tan \alpha_1 = \frac{V^2 - \sqrt{V^4 - g^2 a^2 - 2gV^2 b}}{ga}.$$

p.16

$$(16)(c)(iii) \tan(\alpha_1 + \alpha_2) = \frac{\tan \alpha_1 + \tan \alpha_2}{1 - \tan \alpha_1 \tan \alpha_2} \quad (1)$$

Note: $\tan \alpha_1 + \tan \alpha_2 = \frac{2V^2}{ga} \quad (2)$

$$\begin{aligned} \& \tan \alpha_1 \tan \alpha_2 &= \frac{V^4 (V^4 - ga^2 - 2gV^2b)}{g^2a^2} \\ &= \frac{g^2a^2 + 2gV^2b}{g^2a^2} \\ &= 1 + \frac{2gV^2b}{g^2a^2} \\ &= 1 + \frac{2V^2b}{ga^2} \quad (3) \end{aligned}$$

Sub. (2) & (3) in (1):

$$\begin{aligned} \tan(\alpha_1 + \alpha_2) &= \frac{\frac{2V^2}{ga}}{1 - \left(1 + \frac{2V^2b}{ga^2}\right)} \\ &= \frac{\frac{2V^2}{ga}}{\frac{-2V^2b}{ga^2}} \\ &= \frac{2V^2}{ga} \times \frac{-ga^2}{2V^2b} \\ &= -\frac{a}{b} \end{aligned}$$

As $\tan \theta = \frac{b}{a}$,

$$\begin{aligned} \tan(90^\circ + \theta) &= \tan[180^\circ - (90^\circ - \theta)] \\ &= -\tan(90^\circ - \theta) \\ &= -\frac{1}{\tan \theta} \\ &= -\frac{a}{b} \end{aligned}$$

$$\begin{aligned} \therefore \tan(\alpha_1 + \alpha_2) &= \tan(90^\circ + \theta) \\ \therefore \alpha_1 + \alpha_2 &= 90^\circ + \theta \quad \text{both in } Q_2 \\ \alpha_1 + \alpha_2 &= 90^\circ + \theta \\ \alpha_1 - \theta &= 90^\circ - \alpha_2 \\ \angle BOA &= \angle COY \\ \text{END OF SOLUTIONS!} \end{aligned}$$

(or)

p. 17

(16) (c) (iii) alternative:

 $\tan \alpha_1$ & $\tan \alpha_2$ are solutions to

$$ga^2 \tan^2 \alpha - 2V^2 a \tan \alpha + (ga^2 + 2V^2 b) = 0 \text{ [from p. 15].}$$

By $\beta + \gamma = -\frac{b}{a}$ [sum of roots of quadratic equation]

$$\begin{aligned} \tan \alpha_1 + \tan \alpha_2 &= \frac{2V^2 a}{ga^2} \\ &= \frac{2V^2}{ga} \quad (1) \end{aligned}$$

By $\rho\sigma = -\frac{c}{a}$ [product of roots of quadratic equation]

$$\begin{aligned} \tan \alpha_1 \tan \alpha_2 &= \frac{ga^2 + 2V^2 b}{ga^2} \\ &= 1 + \frac{2V^2 b}{ga^2} \quad (2) \end{aligned}$$

$$\text{Hence } \tan(\alpha_1 + \alpha_2) = \frac{\tan \alpha_1 + \tan \alpha_2}{1 - \tan \alpha_1 \tan \alpha_2} \quad (3)$$

$$\begin{aligned} \text{Sub. (1) \& (2) in (3)} &= \frac{\frac{2V^2}{ga}}{1 - \left(1 + \frac{2V^2 b}{ga^2}\right)} \\ &= \frac{2V^2}{ga} \div -\frac{2V^2 b}{ga^2} \\ &= \frac{2V^2}{ga} \times \frac{ga^2}{-2V^2 b} \\ &= -\frac{a}{b} \end{aligned}$$

$$= \cot \theta$$

$$= \tan(90^\circ + \theta).$$